

Example 11.1

$$X_t = \frac{3}{4} X_{t-1} - \frac{1}{8} X_{t-2} + Z_t$$

$$\left(1 - \frac{3}{4}B + \frac{1}{8}B^2\right) X_t = Z_t$$

$$\Rightarrow \left(1 - \frac{B}{4}\right)\left(1 - \frac{B}{2}\right) X_t = Z_t$$

$$\Rightarrow (1 - \lambda_1 B)(1 - \lambda_2 B) X_t = Z_t$$

$$\text{where } \lambda_1 = \frac{1}{4}, \lambda_2 = \frac{1}{2}.$$

$$\begin{aligned}\hat{X}_{n+1}(n) &= \frac{1}{\left(\frac{1}{4} - \frac{1}{2}\right)} \left\{ \left[\left(\frac{1}{4}\right)^2 - \left(\frac{1}{2}\right)^2 \right] x_n \right. \\ &\quad \left. - \left[\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)^2 - \left(\frac{1}{4}\right)\left(\frac{1}{2}\right)^2 \right] x_{n-1} \right\} \\ &= -4 \left\{ \left(\frac{1}{16} - \frac{1}{4}\right) x_n - \left(\frac{1}{32} - \frac{1}{16}\right) x_{n-1} \right\} \\ &= \frac{3}{4} x_n - \frac{1}{8} x_{n-1}\end{aligned}$$

$$\begin{aligned}\hat{X}_{n+2}(n) &= \frac{1}{\left(\frac{1}{4} - \frac{1}{2}\right)} \left\{ \left[\left(\frac{1}{4}\right)^3 - \left(\frac{1}{2}\right)^3 \right] x_n \right. \\ &\quad \left. - \left[\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)^3 - \left(\frac{1}{4}\right)\left(\frac{1}{2}\right)^3 \right] x_{n-1} \right\} \\ &= \frac{7}{16} x_n - \frac{3}{32} x_{n-1}.\end{aligned}$$